Pre-work · Bone Tissue

Bone Tissue and Microanatomy. Read the Part A chapter first, then work this in order. You come to class ready to use this, not to hear it for the first time.

Module 2 · 1 of 4 | Bring this to class

How to work this pre-work

1 min

The order you do these in matters more than the number of hours you put in. Context has to come before content. Almost every student who struggles with this course did all of the pieces, just not in the sequence that makes them work.

1. **Start with the reading.**
   Open the Part A chapter and read it before you touch anything on this sheet. I have not repeated the chapter here, because this packet is where you put that reading to work.
2. **Organize what you read.**
   In Panel 1, turn the reading into small visuals. Do not copy sentences across. The whole point is that you decide how it fits together.
3. **Draw it.**
   In Panel 2, dump everything you can remember first, then draw it, then go back with a second color and correct yourself. The second color is the part that teaches you something.
4. **Apply it.**
   Work Panel 3 with your notes closed the first time through. Open them afterward, only to check what you got.
5. **Come back to it later in the week**
   , not the same night you drew it. Cover everything and redraw your visuals from memory. Whatever you cannot redraw is the part you have not learned yet, and it is much better to find that out now than on exam day.

Here is why the order works. Reading first gives you the context to hang everything else on. Organizing before you draw forces you to decide what connects to what, which is the thinking part. Drawing before you apply is where you find out what you actually know rather than what only looks familiar. If you jump straight to the application questions you will answer them with your notes open, and you will learn almost nothing from them.

Organize it

2 min

PANEL 1 OF 3

Turn the bone chapter into mini visuals. Seven chunks, each in the shape that fits it.

YOU ONLY NEED FIVE SHAPES

Information has a shape, and if you choose the wrong one your diagram turns into noise. Choose the one that actually fits and you will be able to redraw the whole thing from memory in about twenty seconds. Use a ladder for an ordered series, a chain with arrows for a sequence, a split for a pair you keep confusing, a hub for one structure with parts hanging off it, and a grid when two variables cross.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk takes longer than three minutes you are copying instead of organizing.

1 GRID

The three cartilages

Hyaline, elastic, fibrocartilage down the side. Fibers, texture, and one location across the top. Add a fourth column for the joint each one turns up in.

2 HUB

Gross anatomy of a long bone

One long bone down the middle. Spoke out to diaphysis, epiphysis, metaphysis, medullary cavity, articular cartilage, periosteum, endosteum, epiphyseal line, and both marrows. Draw the bone, do not just list.

3 HUB

The osteon

Central canal at the hub. Spoke out to lamellae, lacunae, canaliculi, perforating canal, interstitial and circumferential lamellae. This is the single most tested structure in the chapter.

4 SPLIT

Compact against spongy bone

One split. Structural unit, what fills the spaces, and where in a long bone each is found. The giveaway on a slide goes on the branch.

5 LADDER

The four bone cells

Three rungs for the lineage, osteogenic to osteoblast to osteocyte, then the osteoclast off to one side because it comes from somewhere else. Write what each does beside its rung.

6 SPLIT

Intramembranous against endochondral ossification

What tissue each starts from, and which bones each builds. Two branches, no more.

7 LADDER

The five zones of the growth plate

Five rungs, resting cartilage at the top by the epiphysis, ossification at the bottom by the diaphysis. Write what the chondrocyte is doing at each rung.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

Draw it

2 min

PANEL 2 OF 3

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

DRAWING 1

A long bone, sectioned

BRAIN DUMP FIRST

Two minutes. Every part of a long bone, and which kind of bone tissue is where.

Draw a long bone cut down its length so the inside shows. Mark diaphysis, both epiphyses, metaphysis, medullary cavity, and where compact and spongy bone each sit. Add periosteum, endosteum, articular cartilage, the epiphyseal line, and label red and yellow marrow where each belongs.

In your notebook, head the page **M2-1 A long bone, sectioned** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can point at any level of your drawing and say which tissue is there and what it is doing.

DRAWING 2

The osteon, in cross section

BRAIN DUMP FIRST

One minute. Every structure in compact bone and how a deep osteocyte gets fed.

Draw one osteon large. Central canal in the middle, concentric lamellae as rings, lacunae between the rings with an osteocyte in each, and canaliculi as fine lines linking them. Add a perforating canal crossing to a neighbouring osteon, and squeeze interstitial lamellae into the gaps between osteons.

In your notebook, head the page **M2-2 The osteon, in cross section** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can trace with your finger the route a nutrient takes from the central canal to the outermost osteocyte.

DRAWING 3

The growth plate, zone by zone

BRAIN DUMP FIRST

One minute. The five zones in order and what happens in each.

Draw a strip of growth plate with the epiphysis at the top and the diaphysis at the bottom. Divide it into five bands. Draw the chondrocytes changing as they descend: resting, then stacked in columns, then swollen, then a calcifying matrix, then bone. Label each zone.

In your notebook, head the page **M2-3 The growth plate, zone by zone** so you can find it later. Correct it in a second color.

YOU KNOW IT WHEN

You can explain why a fracture through this plate threatens future growth, using only your drawing.

Apply it

2 min

PANEL 3 OF 3

First pass with everything closed. Then open the notes in a second color and mark what you missed.

WORKED EXAMPLE

A slide of bone shows rings of matrix around a canal, with small cavities between the rings. Name the tissue and the unit.

Answer. Compact bone. The unit is the osteon.

Reasoning. Work from the structure you can actually see. Rings around a canal means concentric lamellae around a central canal, and that pairing exists only in compact bone. The small cavities are lacunae holding osteocytes. Spongy bone would show none of this, because it has no osteons at all, only trabeculae with marrow in the spaces. So the presence of a central canal is the single feature that settles it.

Set A · Name it

1. The structural unit of compact bone is the ______________________.
2. The cell that resorbs bone matrix is the ______________________.
3. The cartilage capping the epiphysis where it meets another bone is ______________________.
4. The membrane lining the internal bone surfaces is the ______________________.
5. Name the two routes of ossification and one bone built by each.

Set B · Predict it

1. A child fractures through the epiphyseal plate. Predict the long-term consequence and say why.
2. An X-ray shows an epiphyseal line rather than a plate. What does that tell you about the person?
3. Canaliculi are blocked in a region of compact bone. Predict what happens to the osteocytes there.
4. Osteoclast activity outpaces osteoblast activity for years. Predict the effect on the skeleton and name one fracture it makes likely.

Set C · Tell it apart

1. A student says cartilage heals as fast as bone. Correct them in one sentence, using vascularity.
2. Distinguish appositional from interstitial growth in a long bone, by direction and by location.
3. Explain why a twisting force breaks a long bone more readily than a force along its length.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colors, and you can say out loud why each correction was a correction.

Mini visual sheet

2 min

Each frame tells you what to draw in it and what to label. Do them with your notes closed, then open your notes and correct yourself in a second color. Draw straight onto this sheet. If a frame is too small for your handwriting, draw that one in your notebook instead and write the frame number at the top of the page so you can find it again.

M2-1

The three cartilages

GRID

Draw a grid. List the things being compared down the left side, write the features you are comparing them on across the top, and fill in every box.

M2-2

Gross anatomy of a long bone

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M2-3

The osteon

HUB

Put the main structure in the middle of the frame. Draw everything that belongs to it around the outside, label each one, and run a line from each back to the middle.

M2-4

Compact against spongy bone

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

M2-5

The four bone cells

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.

M2-6

Intramembranous against endochondral ossification

SPLIT

Draw a line down the middle of the frame. Put one on the left and the other on the right. Draw and label both, then underneath write the one difference that tells them apart.

M2-7

The five zones of the growth plate

LADDER

Draw the levels in order down the frame, the first one at the top. Label each level and write one line saying how it differs from the level above it.